

SMART DUSTBIN

PROJECT FINAL REPORT

Engineering Research Project

EEE3305

Index No.-: 22ug1-0387

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2025 - March

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1. Introduction

Waste management is important to keep our surroundings clean and safe. Traditional dustbins need to be opened by hand, which can be dirty and inconvenient. In public places, bins often overflow because they are not monitored properly. This can cause pollution and health problems.

To solve this issue, this project introduces a Smart Dustbin that works without touching it. The system uses Arduino Uno, an ultrasonic sensor, and a servo motor to open and close the lid automatically. When a person brings trash close to the bin, the sensor detects it, and the lid opens. After a few seconds, the lid closes by itself. This makes waste disposal easier and more hygienic.

The Smart Dustbin is simple, low-cost, and useful for homes, offices, and public places. In the future, it can be improved with IoT features to track waste levels and solar power for better energy use.

1.1 Project Background

Waste disposal is an essential part of daily life but many people avoid touching public dustbins because they can carry germs. Traditional bins also do not alert when they are full, leading to overflowing waste and environmental pollution.

A Smart Dustbin can help solve these issues by using sensors and automation to make waste disposal cleaner and easier. This project is designed to create a low-cost, automatic dustbin that can be used in homes, offices, and public places. By using an ultrasonic sensor and servo motor, the system ensures that users do not need to touch the bin, improving hygiene and efficiency.

1.2 Objectives

The main objectives of this project are:

1. To design a dustbin that opens and closes automatically using sensors and motors.
2. To reduce direct contact with the dustbin lid and improve hygiene.
3. To make a low-cost and easy-to-build solution for waste disposal.
4. To ensure quick and efficient operation with minimal power consumption.
5. To provide a better waste management system for homes, offices, and public places.

2. Literature Review

In recent years, different smart waste management systems have been developed to improve cleanliness and efficiency. Some of these systems use Internet of Things technology to monitor and manage waste levels. Other projects use RFID sensors and mobile applications to track garbage collection. While these technologies offer good solutions, they can be expensive and complicated to implement.

Several studies have explored the use of ultrasonic sensors in smart dustbins. These sensors help detect objects near the bin and allow for automatic lid operation. Research has shown that using servo motors along with Arduino-based control systems can significantly improve the convenience and hygiene of waste disposal.

However, many smart dustbin projects focus on advanced features like real-time monitoring and GPS tracking, which may not be necessary for small-scale or household use. This project aims to develop a simple, low-cost, and effective smart dustbin that can be easily built and used in various environments. By using Arduino Uno, ultrasonic sensors, and a servo motor, this system provides an affordable and practical solution for smart waste disposal.

3. Methodology

3.1 Hardware Components

- **Arduino Uno**
 - The main controller that processes data and operates the servo motor.
- **Ultrasonic Sensor (HC-SR04)**
 - Detects objects near the bin by measuring distance.
- **Servo Motor**
 - Opens and closes the dustbin lid.
- **Power Supply**
 - Provides power to all components.
- **Dustbin Structure**
 - A regular dustbin modified to fit the electronics.

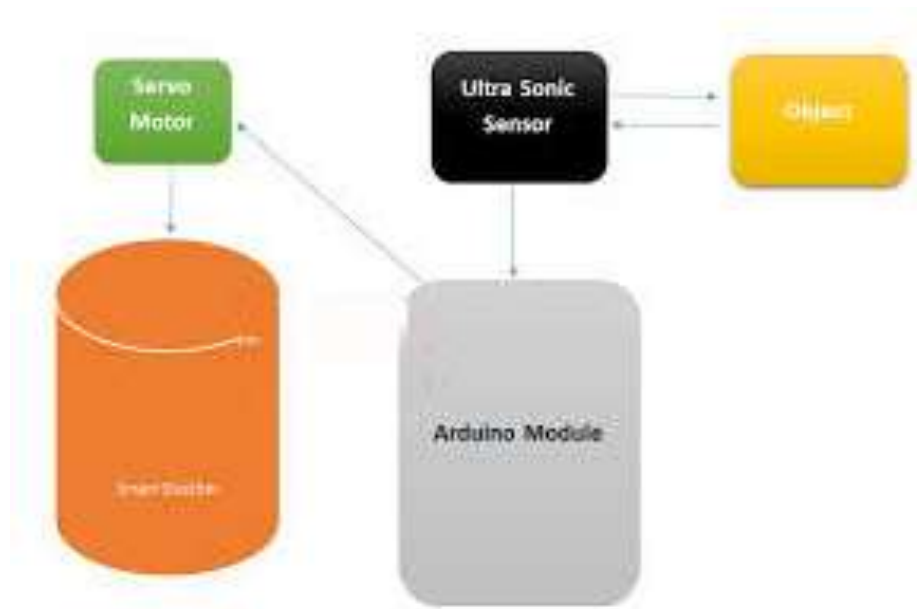


Fig.01 – Component Diagram

3.2 Software Implementation

The Arduino Uno is programmed using the Arduino IDE. The ultrasonic sensor continuously checks for objects. If an object is detected within a certain distance (for example, 10-20 cm), the Arduino sends a signal to the servo motor to open the lid. After a few seconds, the lid closes automatically.

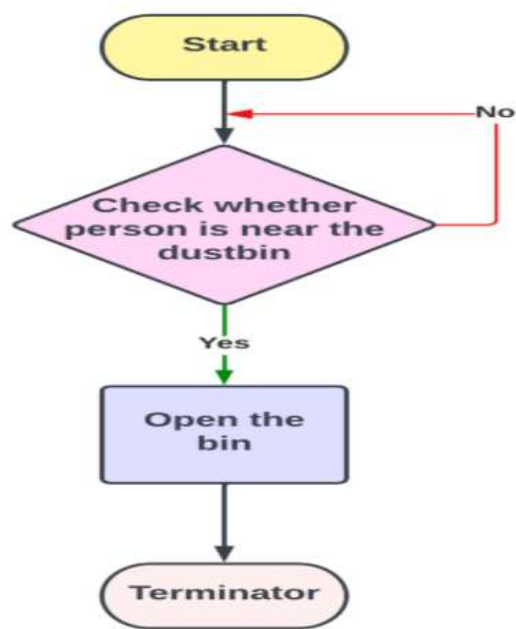


Fig.02 – Flow Chart

4. Working Principle

The Smart Dustbin works using an ultrasonic sensor, an Arduino Uno, and a servo motor. The process follows these steps:

1. The ultrasonic sensor sends out sound waves.
2. These sound waves hit an object (like a hand or trash) and bounce back.
3. The sensor measures the time taken for the waves to return and calculates the distance of the object.
4. If the object is within a set range (e.g., 0-30 cm), the Arduino Uno receives the signal.
5. The Arduino then sends a command to the servo motor to rotate and open the dustbin lid.
6. After a few seconds, the Arduino signals the servo motor to close the lid again.
7. The system keeps running, continuously checking for new objects near the dustbin.

This process is showed in [Fig.02](#) above.

5. Results & Discussion

The Smart Dustbin functioned successfully during testing. When an object was detected within the predefined range, the lid opened automatically and closed after a few seconds. The response time of the system was observed to be 2-10 seconds, making it quick and efficient for daily use. The power consumption was minimal, making it suitable for battery operation or low-power supply environments.

One major advantage of this system is its touch-free operation, which helps in maintaining hygiene and reducing the spread of germs. Additionally, the ultrasonic sensor provided accurate distance measurements, ensuring the lid only opened when necessary.

However, some challenges were encountered during testing. The ultrasonic sensor occasionally gave false readings due to environmental noise, leading to unnecessary lid movements. To solve this, adjustments were made to the sensor position and filtering techniques were used to improve accuracy. Another issue was the servo motor precision, which was resolved by fine-tuning the servo angles in the Arduino code.

Overall, the Smart Dustbin achieved its intended objectives by providing an automatic, hygienic, and cost-effective solution for waste disposal. Future improvements could include IoT integration for real-time waste monitoring and a solar-powered energy system to enhance efficiency.

6. Challenges & Solutions

Challenge	Problem	Solution
1. Sensor Detection Accuracy	The ultrasonic sensor sometimes gave inaccurate readings due to environmental factors (e.g., temperature, obstacles).	Calibrated the sensor, adjusted its range, and tested it in different environments for better accuracy
2. Servo Motor Response	The servo motor occasionally failed to open or close the lid smoothly due to motor placement or insufficient power.	Adjusted the motor's power supply and recalibrated the servo to ensure smooth and consistent movement.
3. Power Consumption	Continuous operation led to high power consumption, which could be problematic for battery-powered systems.	Implemented a low-power sleep mode on the Arduino, which activates the sensor only when movement is detected.
4. Hygiene Maintenance	The automatic lid opened and closed frequently, leading to dirt and bacteria build-up.	Designed the lid with a smooth, easy-to-clean surface and considered adding a UV light or cleaning mechanism.
5. Delay in Lid Response	The lid sometimes had a delayed response after detecting movement, which was frustrating for users.	Reduced the delay time in the Arduino code to allow the lid to open and close faster after detecting motion.
6. Space Constraints	Limited space in public areas sometimes made it difficult to place the Smart Dustbin without obstruction.	Designed a compact version of the Smart Dustbin that could be easily installed in small or crowded spaces.
7. Sensor Sensitivity to Weather Conditions	Outdoor weather conditions like rain or high humidity affected the sensor's performance.	Protected the sensor with a weather-resistant casing to ensure it could operate in various environmental conditions.
8. User Awareness	Some users were unsure how to use the Smart Dustbin, leading to improper disposal of waste.	Added clear instructions on the bin, with diagrams to help users understand its proper use. Future versions could feature voice prompts or a small screen.

Table 01 - Challenges & Solutions

7. Advantages and Disadvantages

Advantages	Disadvantages
1. The bin is touchless, which helps keep it clean and hygienic. 2. It opens automatically, making waste disposal quick and easy. 3. The parts used in the bin are affordable. 4. It can be powered by solar energy, making it eco-friendly. 5. The bin is easy to use with a simple design and clear instructions.	1. The sensor might not work well in environments with noise or rain. 2. It uses power, which can drain the battery if used frequently. 3. The bin may be difficult to place in crowded or small spaces. 4. It needs to be cleaned regularly to stay hygienic. 5. The bin lacks advanced features, such as remote monitoring or IoT capabilities.

Table 02 - Advantages and Disadvantages

8. Conclusion

The Smart Dustbin project successfully demonstrates how technology can improve waste management by automating waste disposal in a hygienic and efficient way. By using an Arduino Uno, an ultrasonic sensor, and a servo motor, the bin opens automatically when it detects nearby movement, reducing the need for manual contact.

Through the course of the project, we faced several challenges, including sensor accuracy, motor response, and power consumption. However, we were able to find effective solutions for each issue, making the system more reliable and user-friendly.

This project has the potential for further improvement, such as adding IoT features for real-time monitoring or using solar power for sustainability. Overall, the Smart Dustbin provides a simple yet effective solution for waste management in homes, offices, and public spaces, helping to reduce pollution and promote hygiene.

9. References

- ▶ <https://www.arduino.cc>
- ▶ [JETIR2205152.pdf](#)
- ▶ <https://www.instructables.com/IoT-Based-Smart-Dustbin/>
- ▶ <https://bigyanproject.com/en/projects/32-contactless-smart-dustbin>
- ▶ <https://projecthub.arduino.cc/>

10. Appendix

10.1 Circuit Diagram

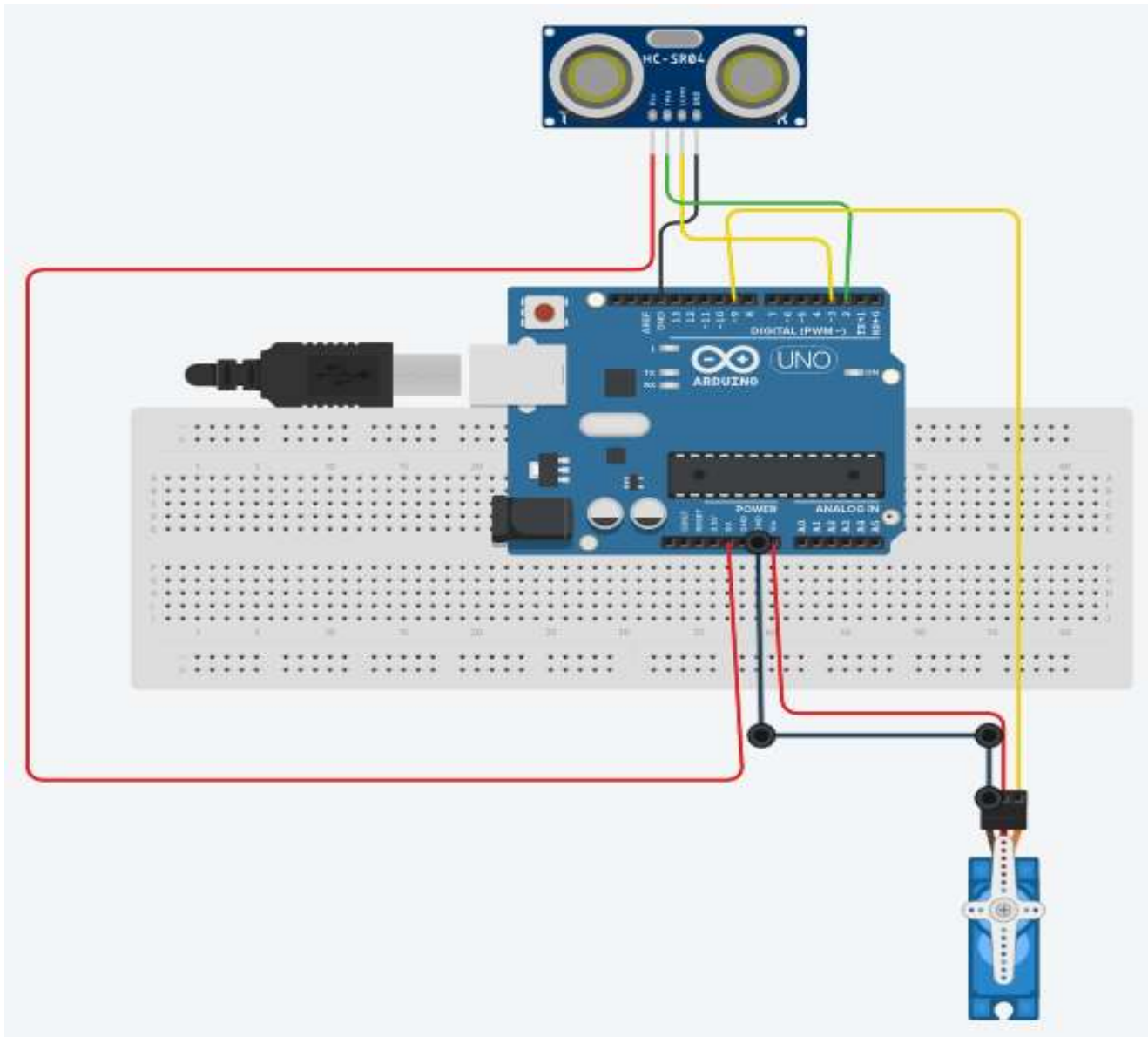


Fig.03 – Circuit Diagram

10.2 Arduino Code

```
#include<Servo.h>

#define trigPin 2
#define echoPin 3

Servo myservo;

void setup() {
  Serial.begin (9600);
  pinMode(trigPin, OUTPUT);

  pinMode(echoPin, INPUT);
  myservo.attach(9);
}

void loop() {
  long duration, distance;
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = (duration / 2) / 29.1;
  if (distance < 30) {

myservo.write(120);
  }
  else {
    myservo.write(0);
  }
  delay(500);
}
```